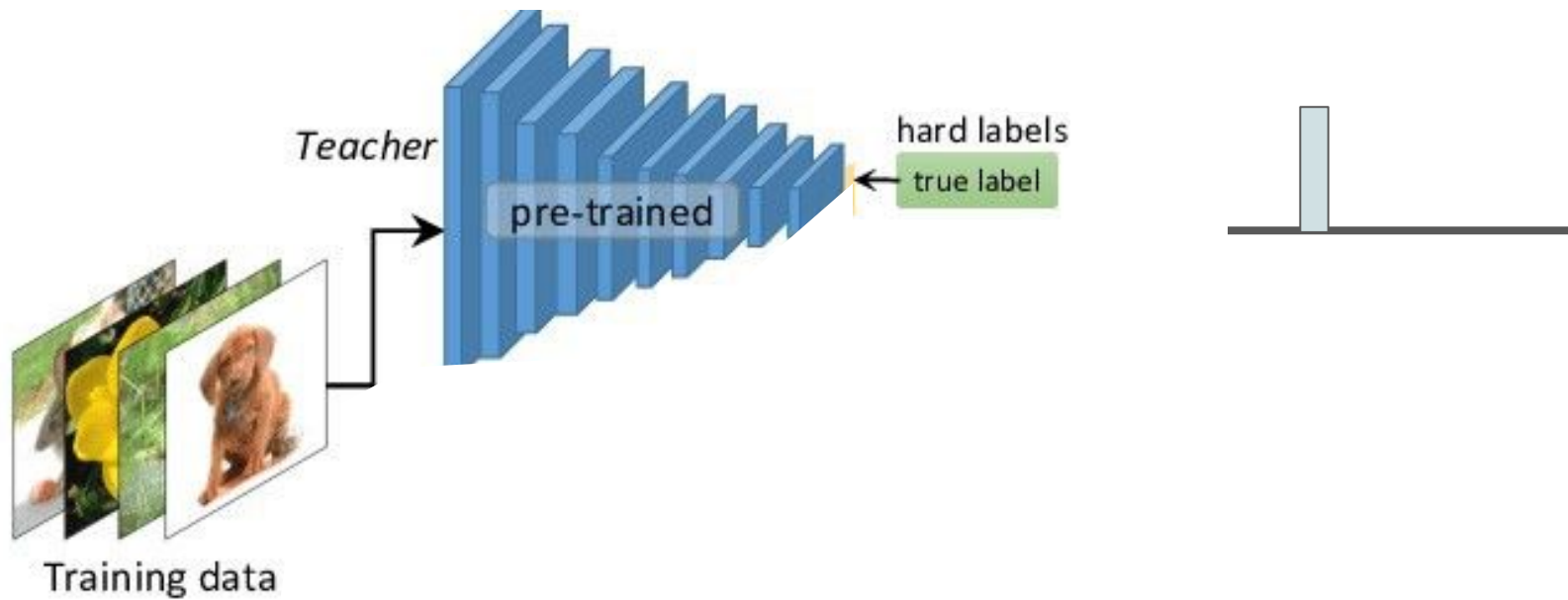


Knowledge Distillation

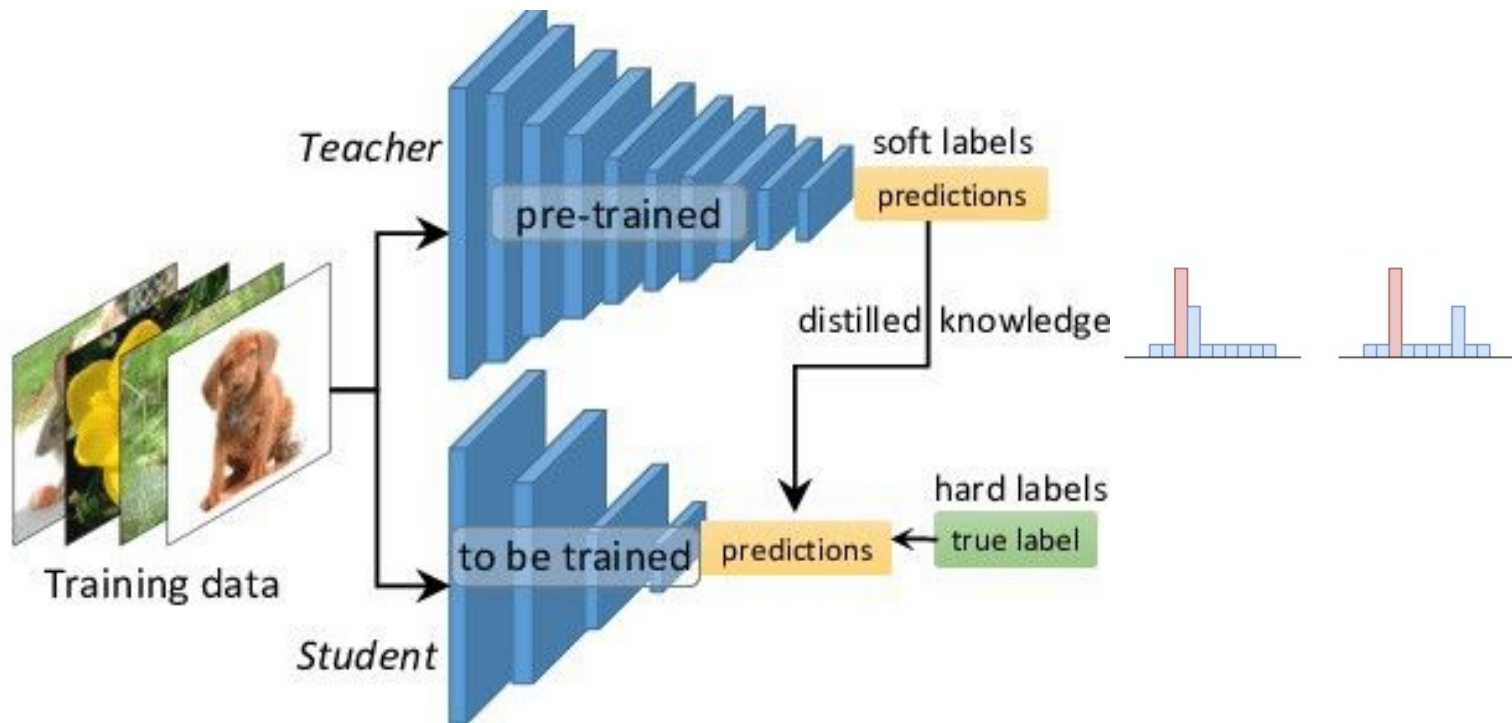
<https://arxiv.org/pdf/2006.05525.pdf>

<https://arxiv.org/pdf/1905.05901.pdf>

<https://arxiv.org/abs/2503.12067>

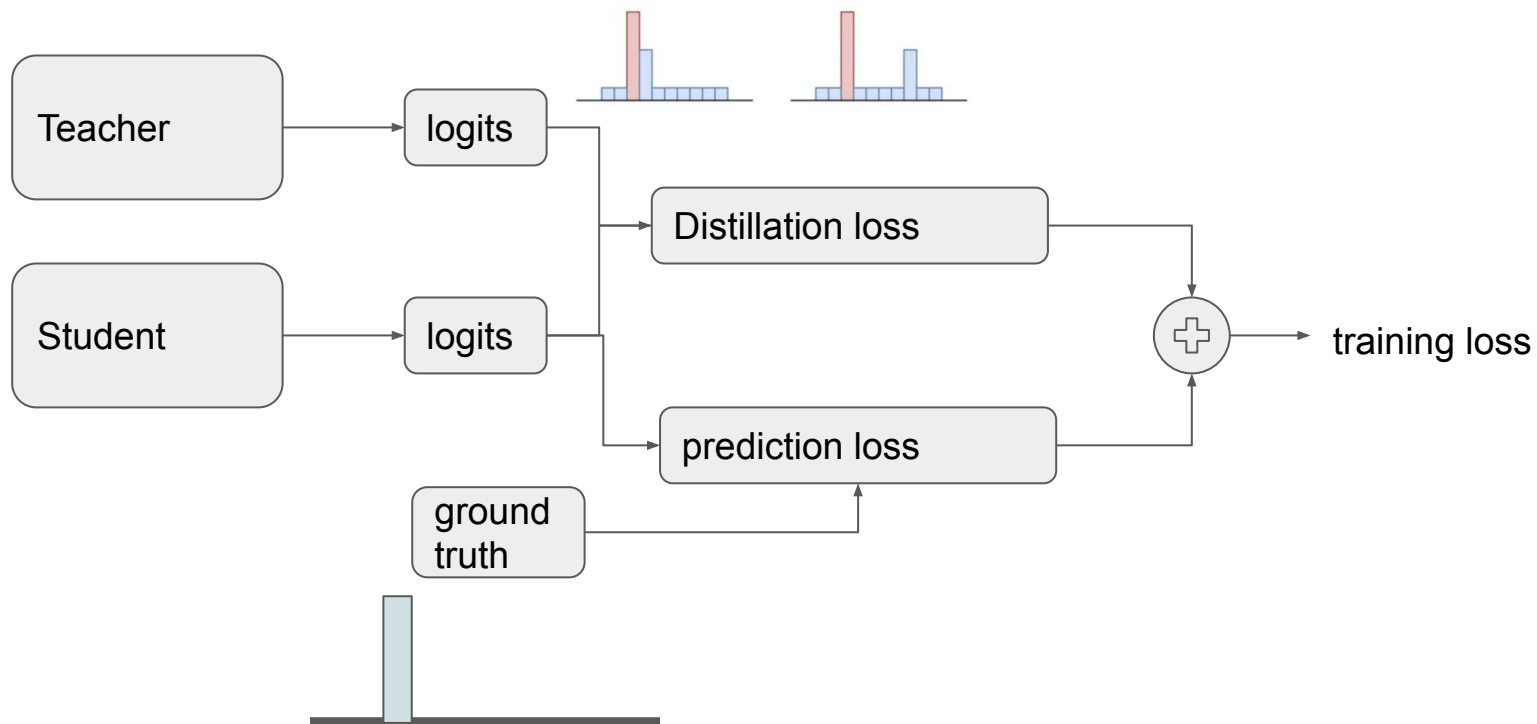


Pretrain the teacher's model by using ground truth(hard labels)



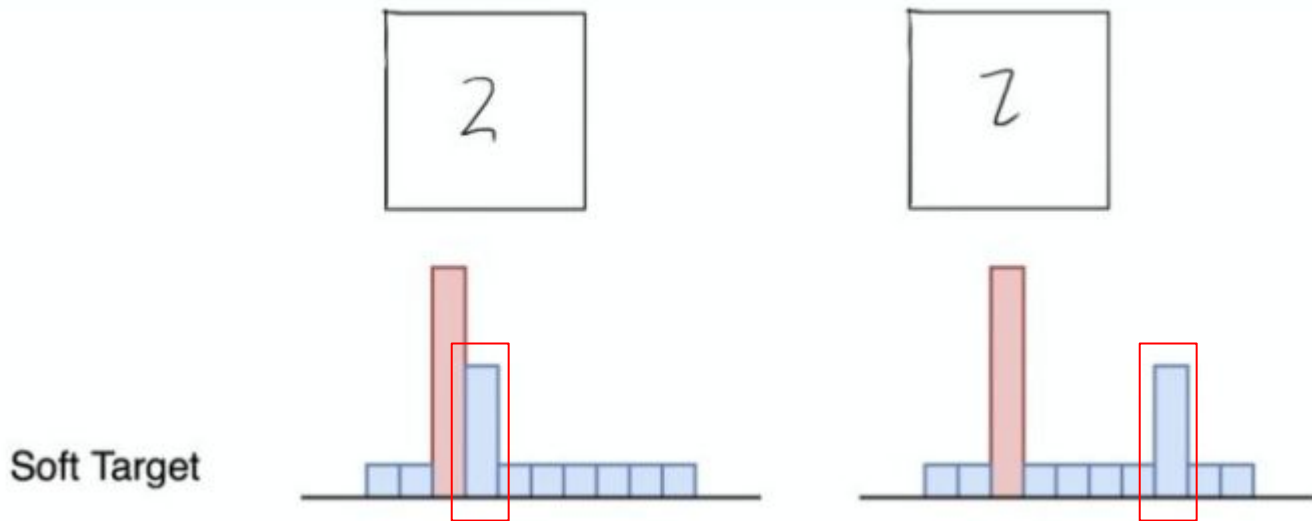
The soft label contains not only the ground truth, but also the less importance features.

Teacher student model

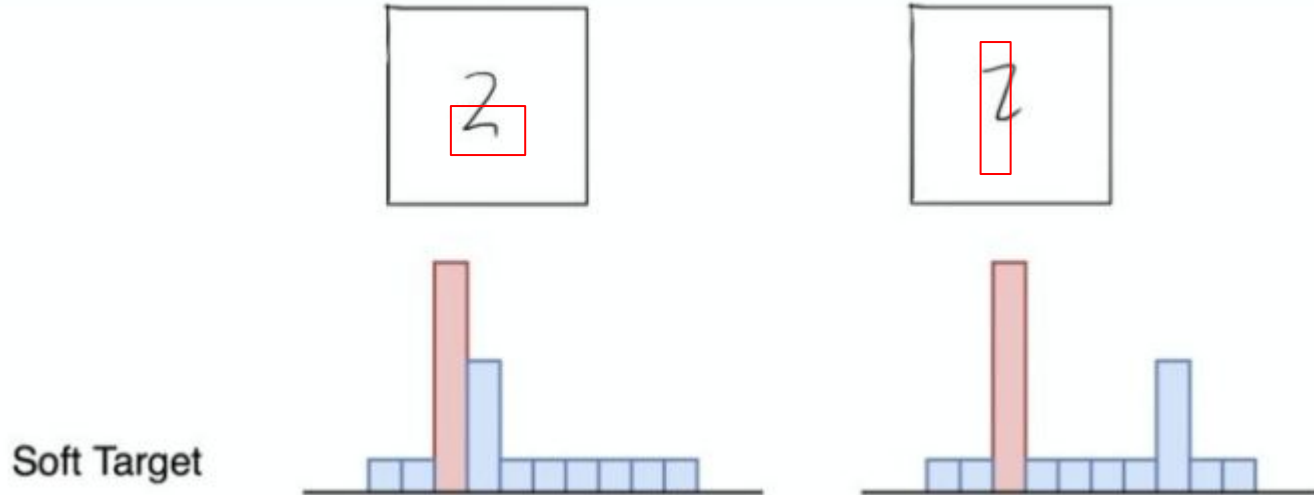


What's soft label?

- We use the logits generated from the teacher's network as the ground truth of the student network to calculate the distillation loss.
- This is similar to use a “smoothed” label.
 - The ground truth is not $[0\ 0\ 0\ 1\ 0\ 0\ \dots]$ any more.



The background information are not all equal importance.



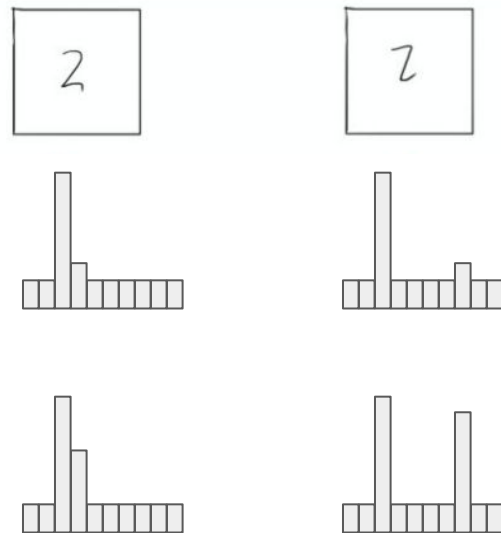


We can not tell the difference between toy garbage car and carrot from ground truth label.



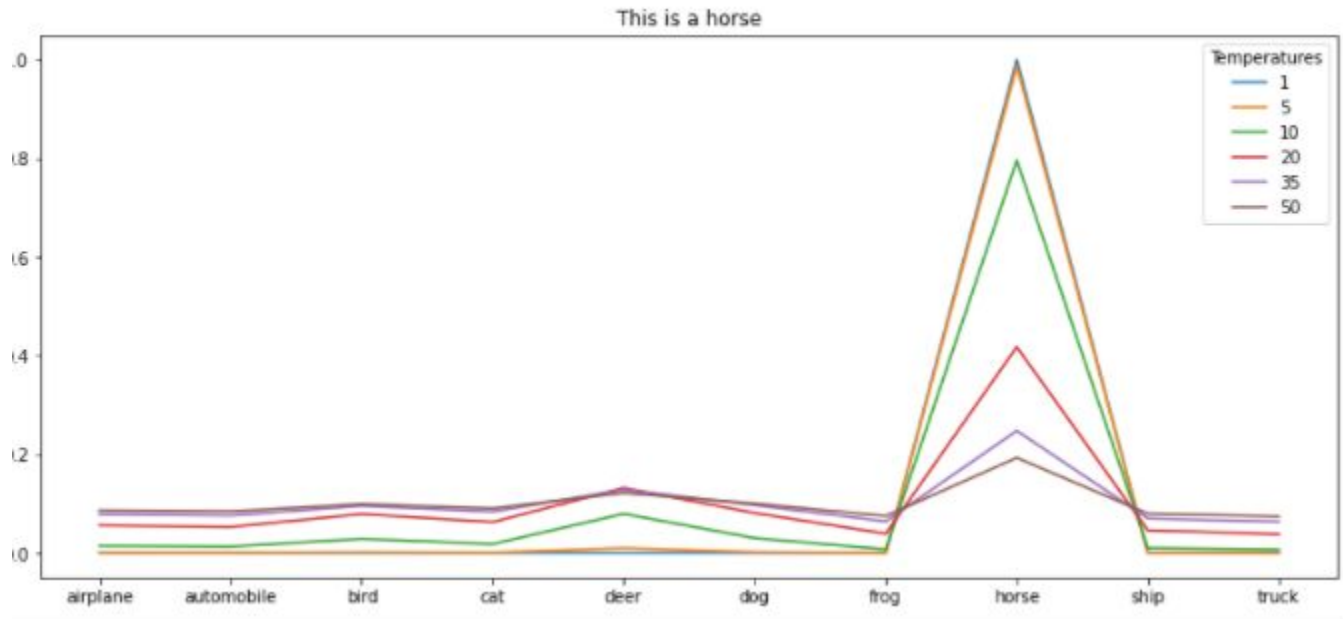
Teacher's model can capture more detailed feature

- A powerful feature extractor in teacher's can capture subtle information.
- If we can force the student network emulate the behaviour of the teacher network, it may get better results.

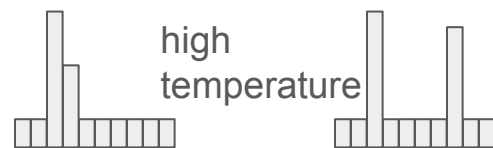
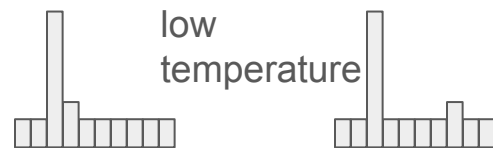


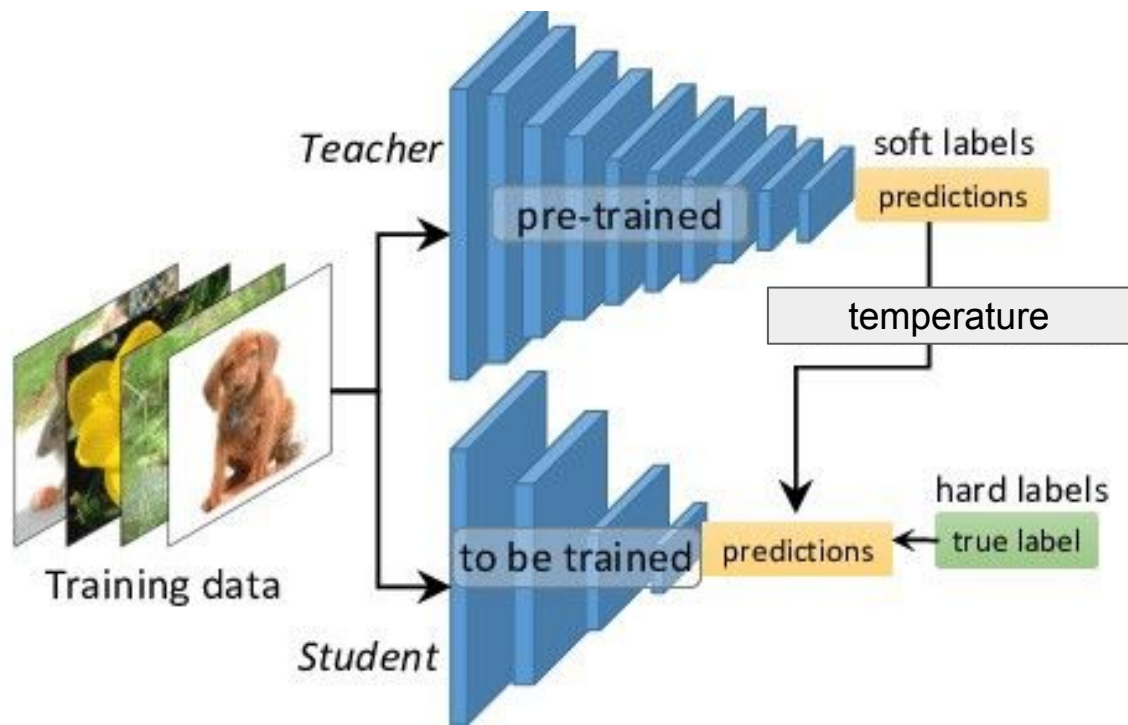
- However, the probability distribution is often very sharp.
- We might soften the distribution.

Temperature in softmax



$$q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$



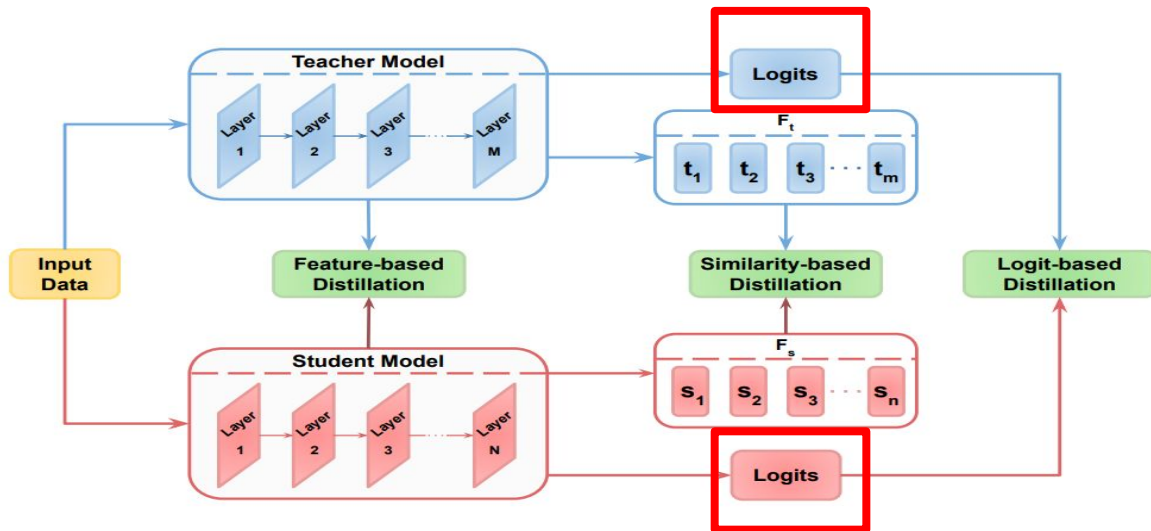


Knowledge Distillation: key components

- Knowledge: What to transfer?
- Distillation Algorithm: How to transfer?
- teacher-student architecture: Who to participate?

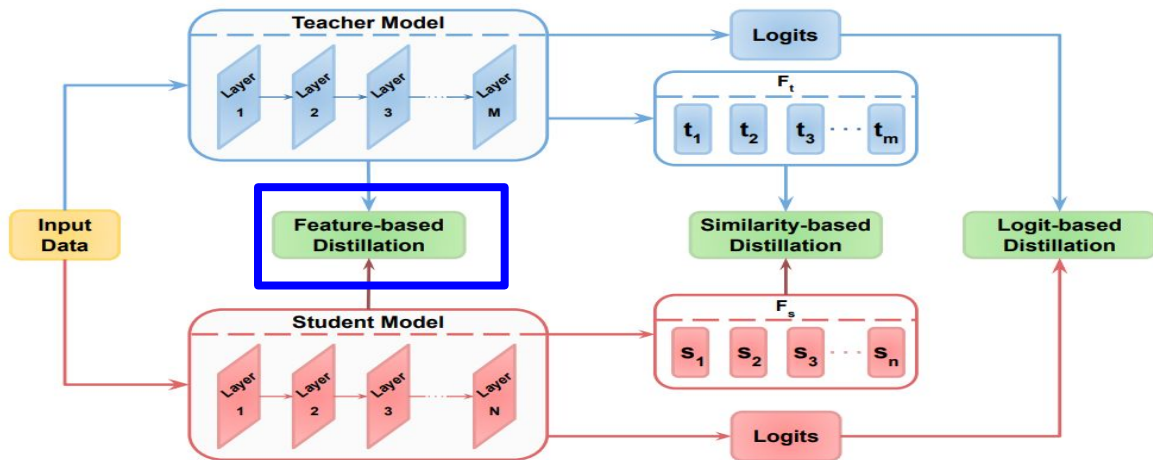
What can we transfered?

- **logits**
- feature maps
- weights
- attention map(heatmap)
- relation between different tensors
- all of them



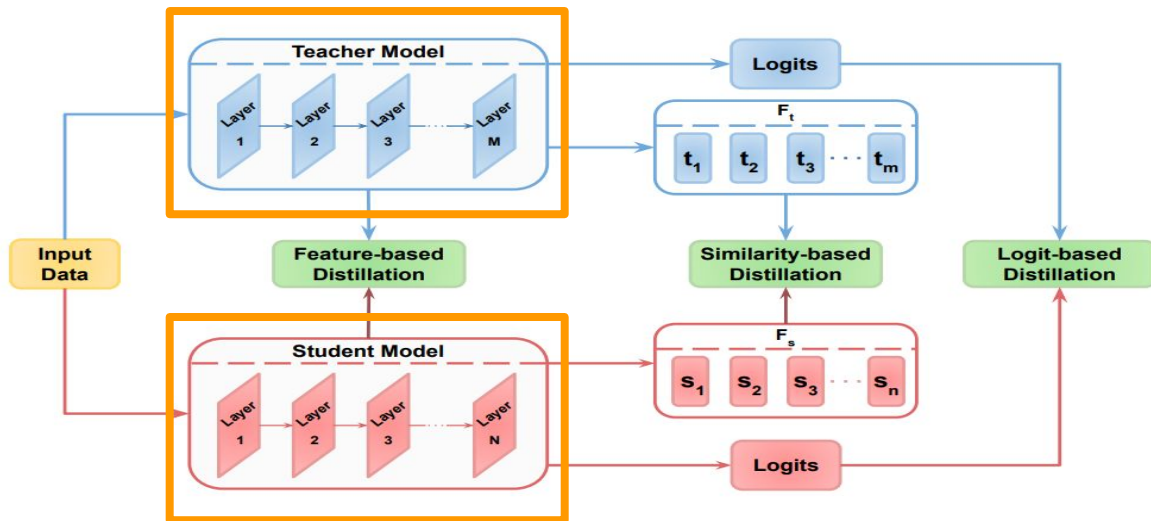
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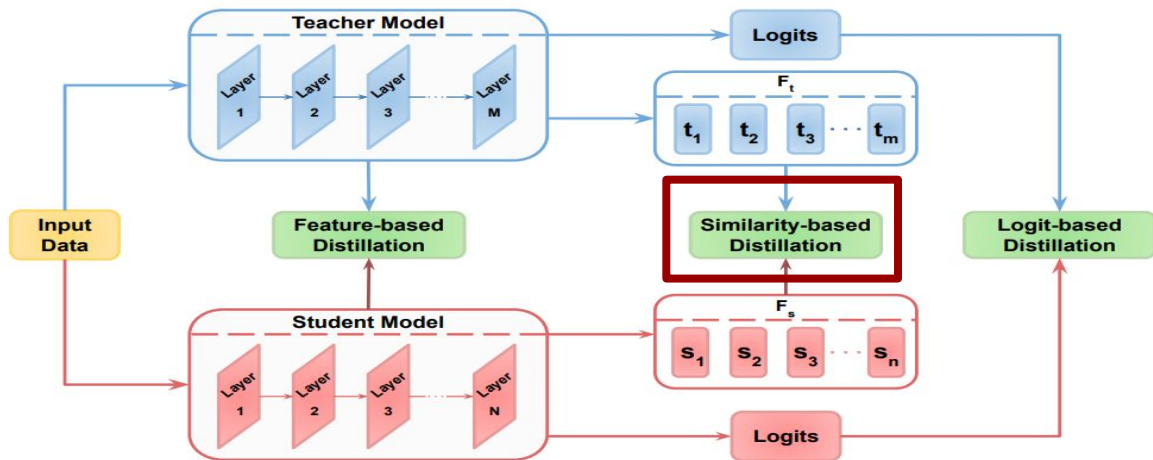
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- all of them



What can we transfered?

- logits
- feature maps
- weights
- attention map(heatmap)
- relation between different tensors
- all of them



attention map

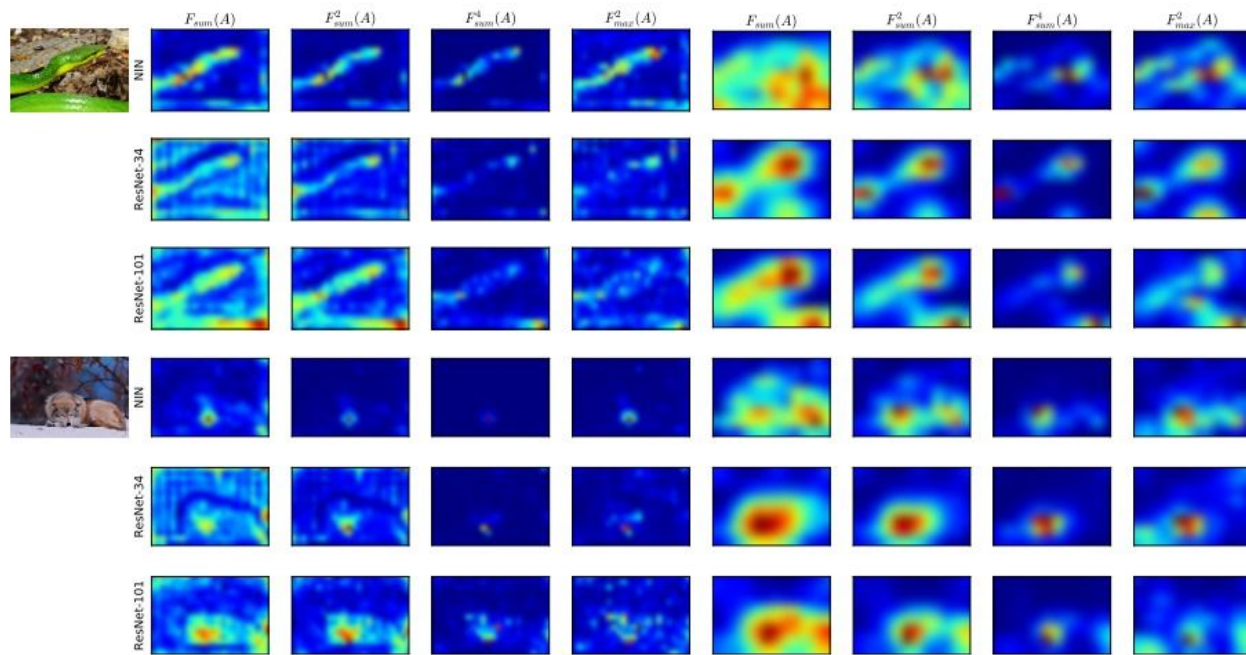
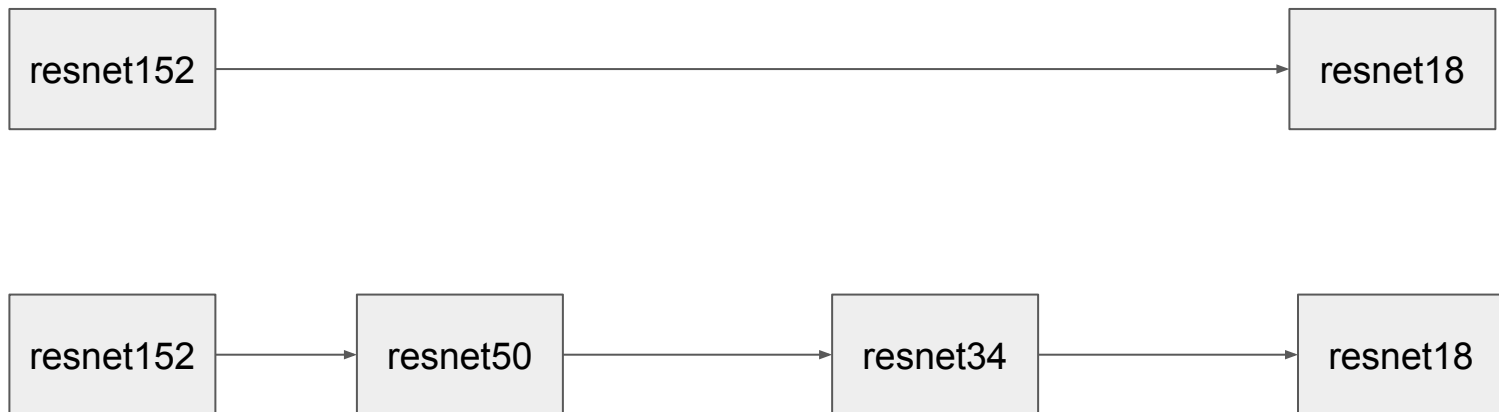


Figure 4: Activation attention maps for various ImageNet networks: Network-In-Network (62% top-1 val accuracy), ResNet-34 (73% top-1 val accuracy), ResNet-101 (77.3% top-1 val accuracy). Left part: mid-level activations, right part: top-level pre-softmax activations

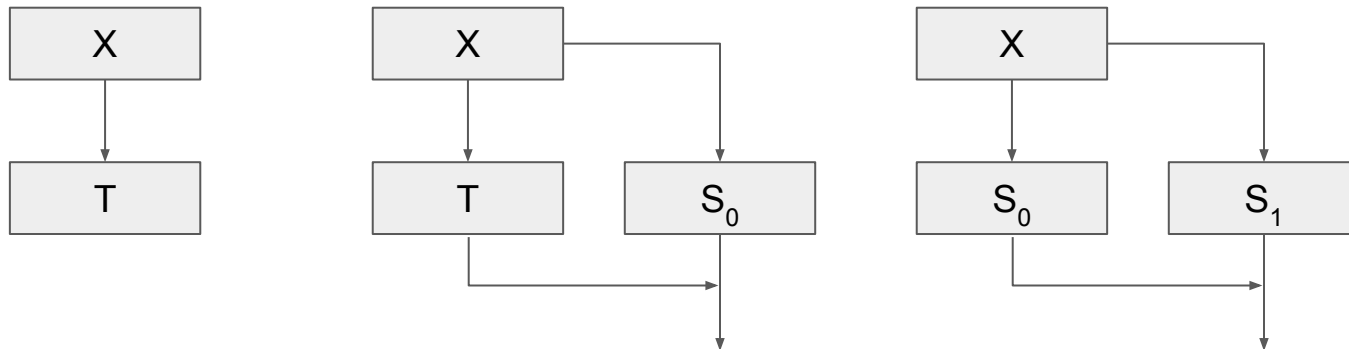
relation-based KD



Multiple teachers: TA

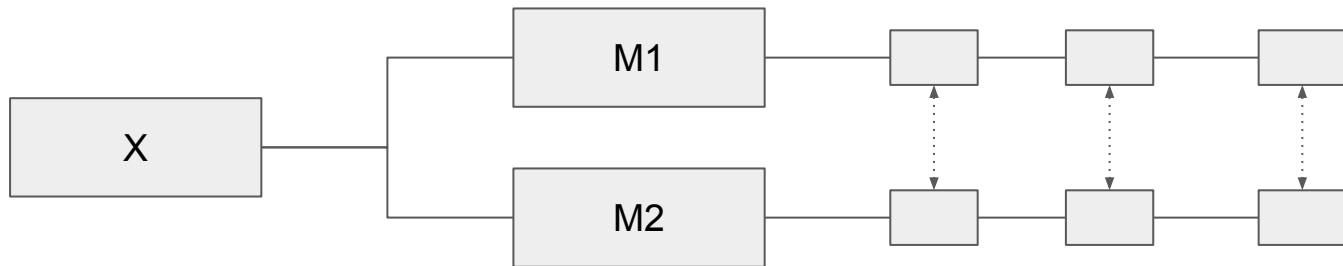


self-distillation



Learn from the previous generation of itself.

online distillation



Two or more models learning at the same time.
They help each other to enhance the performance.

Deep Mutual Learning result

Dataset	Network Types		Independent		1 distills 2	DML	
	Net1	Net 2	Net 1	Net 2	Net 2	Net 1	Net 2
CIFAR-100	WRN-28-10	ResNet-32	78.69	68.99	69.48	78.96	70.73
	MobilNet	ResNet-32	73.65	68.99	69.12	76.13	71.10
Market-1501	Inception V1	MobileNet	65.26	46.07	49.11	65.34	52.87
	MobileNet	MobileNet	46.07	46.07	45.16	52.95	51.26

Table 4: Comparison with distillation on CIFAR-100 (Top-1 accuracy (%)) and Market-1501 dataset (mAP (%))

Combination:

- There are lots of possibilities.
 - Lots of the combination can future improve the performance.

Recent advanced inKD

- [A Comprehensive Survey on Knowledge Distillation](#)
- [DeepKD: A Deeply Decoupled and Denoised Knowledge Distillation Trainer](#)
 - KD works better when you don't trust everything the teacher says
- [TopKD: Top-scaled Knowledge Distillation](#)
 - KD should focus on the small subset of teacher outputs that truly matter
- [Evidential Knowledge Distillation](#)
 - Future KD will incorporate uncertainty, not just probabilities.
- [Neural Collapse-Inspired Knowledge Distillation](#)
 - KD that matches geometric structure—not raw features—can be more robust and interpretable.
- [Feature Alignment–Based Knowledge Distillation for LLMs \(2024/2025\)](#)
 - KD for LLMs must be multi-level — logits alone are not enough.
- [All You Need in Knowledge Distillation Is a Tailored Loss](#)
- [Knowledge Distillation for Language Models](#)
- [Gradient-Guided Knowledge Distillation for Object Detectors](#)